
EXECUTIVE ANALYTICS REPORT

Government Procurement Intelligence Dashboard

An Analysis of Government e-Marketplace (GeM) Procurement Data

Table of Contents

Table of Contents.....	1
1. Executive Summary.....	3
Key Figures at a Glance.....	3
2. Business Context.....	4
3. Project Objectives.....	4
4. Dataset Overview.....	5
4.1 Source.....	5
4.2 Dataset Characteristics.....	5
4.3 Core Attributes.....	5
5. Data Preparation & Methodology.....	6
5.1 Data Cleaning.....	6
5.2 Feature Engineering.....	6
5.3 Methodology Summary.....	6
6. Key Performance Indicators.....	7
Total Procurement Value.....	7
Total Amount Paid.....	7
Total Pending Payment.....	7
Payment Completion Rate.....	7
Total Contracts.....	7
7. Dashboard Overview.....	8
7.1 Design Principles.....	8
7.2 Layout Components.....	8
8. Procurement Analysis.....	9
8.1 Ministry-Level Concentration.....	9
8.2 Organizational Distribution.....	9

8.3 Order Size Segmentation	10
9. Payment Analysis	11
9.1 Payment Completion	11
9.2 Bill Status Distribution	11
9.3 Outstanding Liability Concentration	11
10. Organizational Performance.....	12
11. Buyer Performance	13
11.1 Procurement Concentration Among Buyers	13
11.2 Workload and Payment Responsibility	13
12. Risk Assessment	14
12.1 High-Risk Contracts	14
12.2 Mega-Order Exposure.....	14
12.3 Current Limitations of the Risk Model	14
13. Executive Insights.....	15
14. Strategic Recommendations.....	16
15. Limitations.....	16
16. Future Enhancements	17
17. Technical Architecture	18
17.1 Excel Layer.....	18
17.2 SQL Server Layer	18
18. Conclusion	19

1. Executive Summary

Government organizations execute thousands of procurement transactions across ministries, departments, and field offices every year, generating high-volume, high-complexity data that is rarely consolidated into a single point of reference. Without a centralized view, decision-makers struggle to answer even fundamental questions about where money is being spent, how efficiently it is being paid out, and which entities carry the greatest operational or financial exposure.

This report documents the design, methodology, and findings of the Government Procurement Intelligence Dashboard, an end-to-end analytics solution built on Government e-Marketplace (GeM) procurement data. The project converts a raw, transaction-level export of 46,738 procurement records into a governed, decision-ready reporting layer that surfaces procurement value, payment performance, organizational concentration, buyer activity, and payment risk in a single executive view.

The analysis shows that government procurement expenditure is highly concentrated: a small number of ministries, organizations, and buyers account for a disproportionate share of total spend. Payment completion, while strong in aggregate, is uneven across organizations, with a distinct subset of entities carrying the majority of outstanding liabilities. These patterns form the basis of the strategic recommendations presented later in this report.

Key Figures at a Glance

Metric	Value
Total Procurement Value	₹ 156.8 Cr
Total Amount Paid	₹ 132.2 Cr
Total Outstanding Payment	₹ 24.6 Cr
Payment Completion Rate	84.2%
Total Procurement Contracts	46,738
Average Contract Value	₹ 33,554

Figure 1.1 — Executive KPI summary, replace with live figures exported from the dashboard.

These figures represent the current snapshot of the dataset as of the reporting date and are refreshed each time the underlying GeM export is reprocessed through the data preparation pipeline described in Section 5.

2. Business Context

The Government e-Marketplace (GeM) is the primary digital procurement platform used by Indian government ministries, departments, autonomous bodies, and public sector organizations to purchase goods and services. Every transaction on the platform generates a procurement record capturing the buying entity, the vendor relationship, the contract value, and the payment lifecycle from order placement through invoice settlement.

At the scale government procurement operates, this data is generated far faster than it can be manually reviewed. Procurement officers, finance teams, and oversight bodies are typically left to work from static spreadsheets or fragmented departmental reports, which makes it difficult to answer questions that matter most to leadership:

- Which ministries and organizations account for the largest share of procurement expenditure?
- How much government liability remains unpaid at any given point in time?
- Are there structural bottlenecks that consistently delay payment settlement?
- Is spending concentrated among a small number of buyers or organizations, and what risk does that concentration represent?
- How has procurement activity trended over recent years, and where is it heading?

This project addresses that gap directly. Rather than treating procurement data as a static export to be filed away, it treats it as an operational asset that should be monitored, interrogated, and acted upon — the same standard applied to procurement analytics functions inside large private-sector organizations and consulting practices.

3. Project Objectives

The project was scoped around a small set of outcomes that a procurement leadership team would realistically expect from an analytics deliverable:

- Develop a single, centralized dashboard consolidating procurement activity across all ministries, departments, and organizations represented in the dataset.
- Quantify procurement expenditure and payment status at both the aggregate and organizational level.
- Monitor payment completion and surface outstanding liabilities before they become material risks.
- Track procurement activity over time to identify trend shifts and seasonality.
- Segment procurement contracts by size to distinguish routine purchasing from high-value, high-scrutiny transactions.
- Establish a rule-based view of payment risk that flags organizations and contracts warranting closer review.
- Build a technical foundation — clean, well-typed, documented data — that can be extended into SQL Server analysis and, subsequently, Power BI and Python-based automation.

4. Dataset Overview

4.1 Source

The dataset originates from the Government e-Marketplace (GeM) procurement platform and reflects a snapshot export of procurement records as of the reporting date. It is transaction-level data — each row represents a single procurement contract rather than an aggregated summary.

4.2 Dataset Characteristics

Metric	Value
Total Records	46,738
Total Columns / Features	29+
Granularity	One row per procurement contract
Data Type	Transaction-level procurement data
Coverage	Central government ministries, departments & organizations
Snapshot Date	06 July 2026

4.3 Core Attributes

The dataset contains organizational, financial, and temporal fields. The most analytically significant fields are summarized below.

Field	Description
Ministry / Department	The government ministry and department responsible for the procurement
Organization / Office Name	The specific purchasing entity and office
Contract Number	Unique identifier for each procurement contract
Order Amount	Total contract value in INR
Amount Paid / Due Payment	Amount settled to date and amount outstanding
Buyer Name / Designation	The procurement officer responsible for the transaction
Order Date / Delivery Date / Invoice Date	Key milestones in the procurement lifecycle
Bill Status	Current stage of the payment / billing process

5. Data Preparation & Methodology

Raw procurement exports are rarely analysis-ready. Before any KPI or visualization could be trusted, the dataset was put through a structured cleaning and feature engineering process in Excel, designed to mirror the discipline of a professional data preparation pipeline.

5.1 Data Cleaning

- Verified the uniqueness and completeness of contract numbers to confirm one row per procurement transaction.
- Standardized monetary fields into consistent numeric formats, removing text artifacts and inconsistent currency notation.
- Converted date fields (Order Date, Delivery Date, Invoice Date) into true date data types to enable time-based analysis.
- Organized the dataset into a structured Excel Table to support dynamic pivoting and slicer-based filtering.
- Profiled the dataset for missing or null values across key organizational and financial fields.

5.2 Feature Engineering

A set of derived fields was engineered specifically to support the KPIs and segmentation used throughout the dashboard:

Derived Field	Purpose
Amount Paid	Calculated from Order Amount and Due Payment to represent settled value
Outstanding Payment %	Proportion of contract value still unpaid
Payment Completion %	Proportion of contract value already settled
Order Size Category	Segments contracts into Small, Medium, Large, and Mega bands
Order Year / Month / Quarter	Enables trend and seasonality analysis
Payment Risk	Rule-based classification flagging high-exposure contracts

5.3 Methodology Summary

The overall approach follows a standard analytics lifecycle: raw data is cleaned and validated, engineered features are layered on top, the resulting structured table feeds an interactive Excel dashboard, and the same structured dataset is designed to be loaded into SQL Server for deeper, query-driven analysis. This mirrors how procurement and finance analytics teams typically operate — a governed dataset feeding multiple downstream consumption layers rather than a single, disposable spreadsheet.

6. Key Performance Indicators

The dashboard is built around a small number of KPIs, deliberately limited so that a first-time viewer can understand procurement health within roughly fifteen seconds of opening the report.

Total Procurement Value

The total monetary value of all procurement contracts captured in the dataset. This is the primary measure of government purchasing activity.

Total Amount Paid

The cumulative value of procurement contracts that have been fully or partially settled. This measures actual cash outflow against total commitments.

Total Pending Payment

The outstanding liability across all procurement contracts — the government's unpaid financial exposure at the reporting date.

Payment Completion Rate

The proportion of total procurement value that has already been paid, calculated as Amount Paid divided by Total Procurement Value. This is the single most important efficiency metric on the dashboard, as it converts a large monetary figure into an easily benchmarked percentage.

Total Contracts

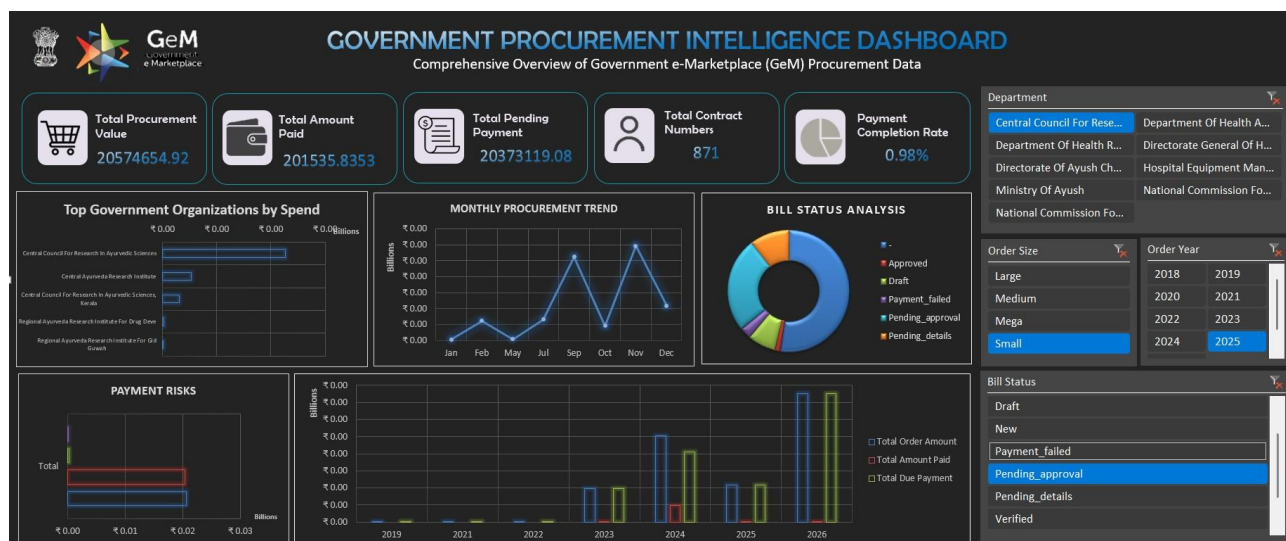
The total count of procurement transactions, used alongside Average Contract Value to describe the shape of government purchasing activity — whether spend is driven by a large number of small purchases or a smaller number of high-value contracts.

7. Dashboard Overview

The dashboard was designed as a single-page, executive-facing report rather than a multi-tab workbook, on the premise that most viewers — recruiters, hiring managers, and time-constrained executives alike — will spend under a minute engaging with it. Every element on the page was chosen to support a fifteen-second read of procurement health.

7.1 Design Principles

- Full-HD (1920×1080) layout optimized for a single screen, with no scrolling required.
- A restrained dark navy theme (background '12, 26, 38', cards '28, 39, 54') with a single accent blue, used consistently for procurement-related visuals, echoing modern BI tools such as Power BI and Microsoft Fabric.
- KPI cards positioned at the top of the page as the first thing a viewer sees, with large, Crore-formatted numbers and short subtitles for context.
- No more than five slicers (Ministry, Department, Order Year, Bill Status, Order Size), positioned in a single horizontal row to maximize space available for charts.

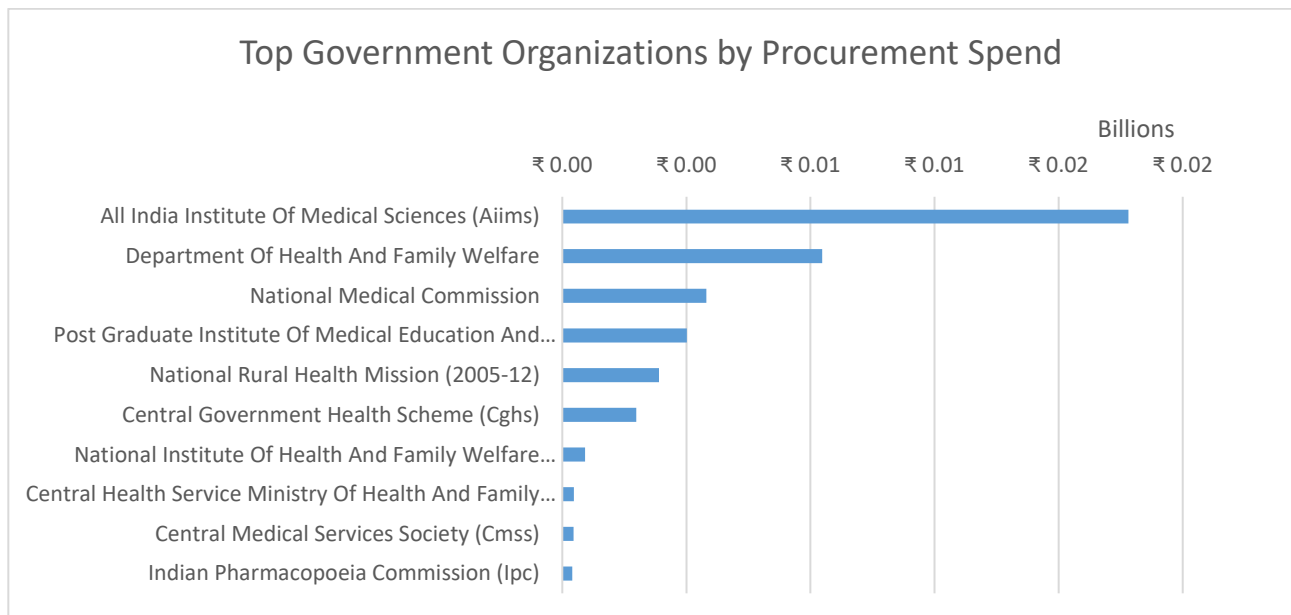
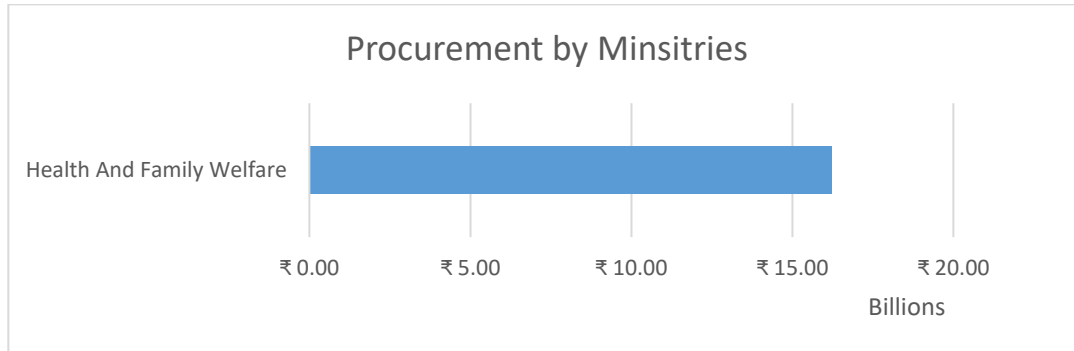


7.2 Layout Components

Section	Purpose
Executive KPI Cards	Total Procurement, Amount Paid, Pending Payment, Contracts, Completion %
Ministry / Organization Analysis	Ranks top-spending ministries and organizations
Monthly / Yearly Procurement Trend	Tracks procurement activity over time
Bill Status Breakdown	Distribution of contracts across billing stages
Payment Risk Panel	Highlights high-risk, high-exposure contracts
Order Size Distribution	Segments contracts into Small, Medium, Large, Mega
Interactive Slicers	Ministry, Department, Order Year, Bill Status, Order Size

8. Procurement Analysis

Procurement expenditure across the dataset is not evenly distributed. A relatively small number of ministries and organizations are responsible for the majority of total spend, a pattern consistent with how large, decentralized purchasing bodies typically operate.



8.1 Ministry-Level Concentration

Ranking ministries by total procurement value shows a clear top tier of high-spend ministries that together account for a disproportionate share of the total. This concentration is a normal feature of large procurement ecosystems, but it also means that dashboard monitoring — and any future risk controls — deliver the greatest value when focused on this top tier rather than spread evenly across all ministries.

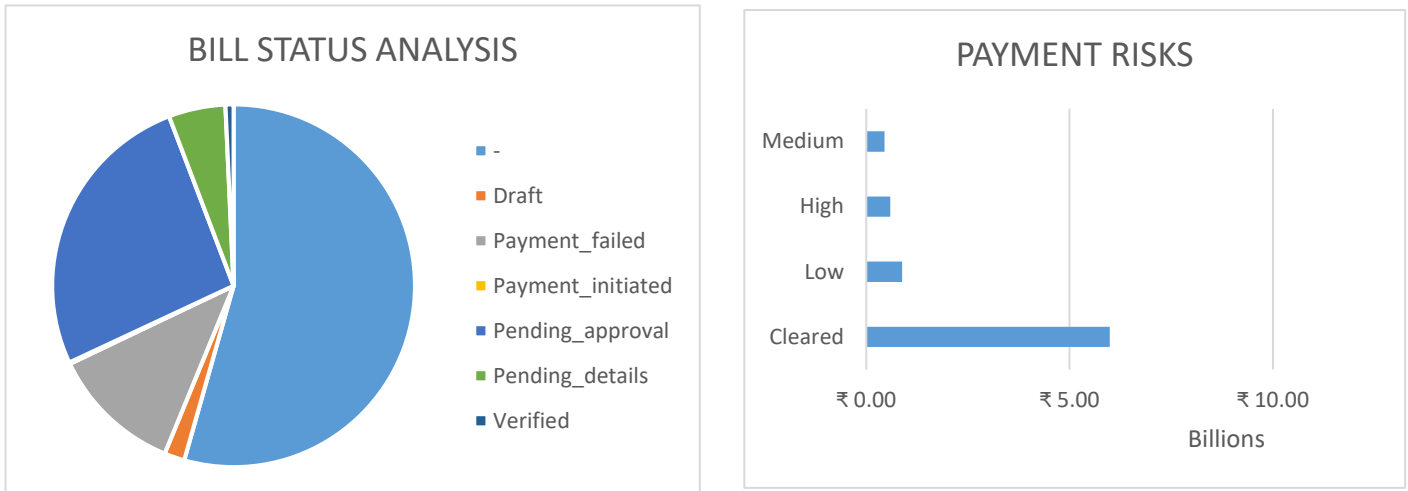
8.2 Organizational Distribution

At the organization level, the same pattern repeats at a finer grain: a limited set of organizations within the top ministries drive most of the procurement value attributed to their parent ministry. This suggests that procurement oversight would be most efficient if targeted at the organization level rather than the ministry level alone.

8.3 Order Size Segmentation

Segmenting contracts into Small, Medium, Large, and Mega bands shows that the majority of individual contracts fall into the Small and Medium categories by count, while a much smaller number of Mega contracts contribute a disproportionately large share of total procurement value. This is a classic Pareto-style distribution and is explored further in the advanced SQL analysis described in Section 17.

9. Payment Analysis



9.1 Payment Completion

At an aggregate level, payment completion across the dataset stands at 84.2%, indicating that the substantial majority of procurement value has been settled. This is a healthy headline figure, but it masks meaningful variation at the organizational level.

9.2 Bill Status Distribution

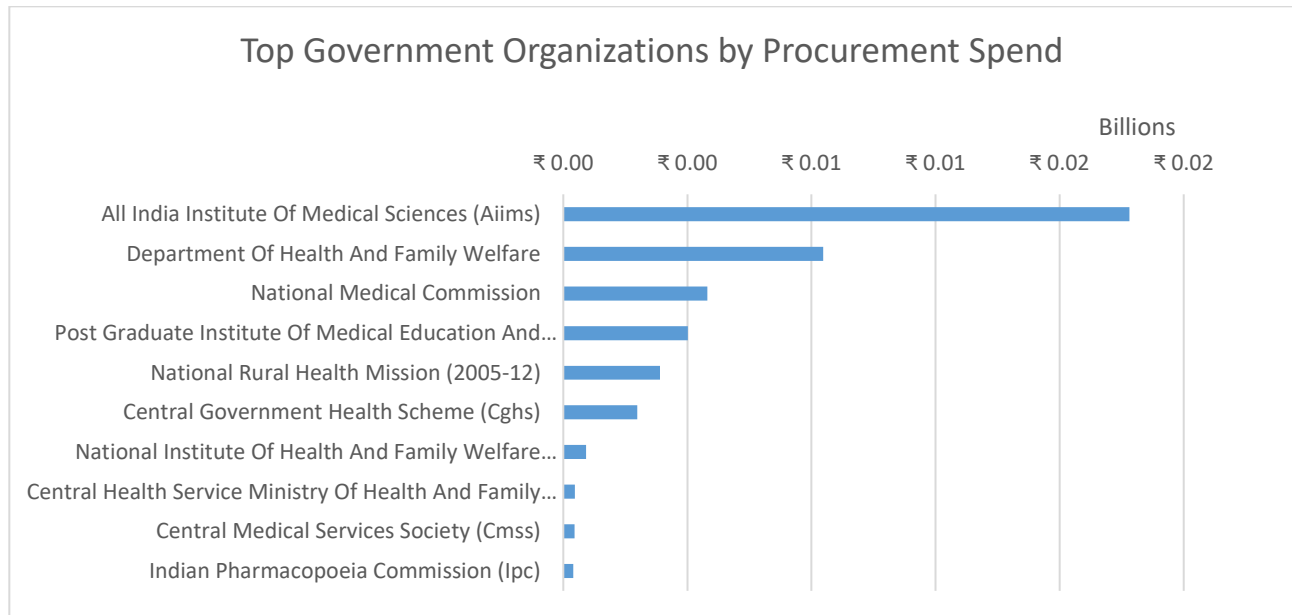
Breaking contracts down by bill status shows that most contracts are in a settled or near-settled state, with a smaller but material share sitting in earlier stages of the billing pipeline. This distribution is a useful early-warning signal: a growing share of contracts stuck in early billing stages would indicate an emerging bottleneck before it shows up as a spike in outstanding payment value.

9.3 Outstanding Liability Concentration

Outstanding payments, like procurement value itself, are concentrated rather than evenly spread. A limited number of organizations account for a disproportionate share of the ₹24.6 Cr in total pending payment, which means remediation efforts focused on this subset of organizations would resolve a large share of total outstanding liability.

10. Organizational Performance

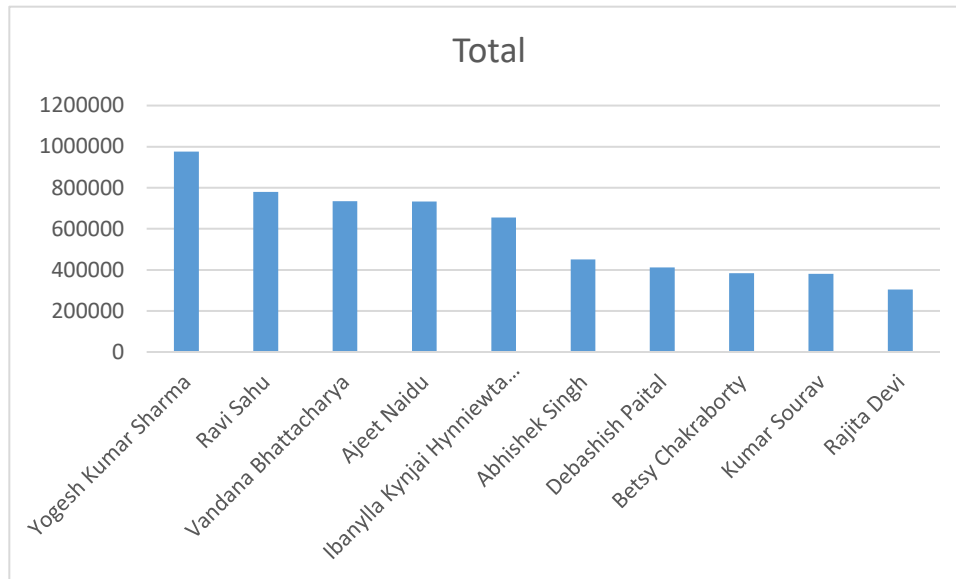
Viewing procurement through an organizational lens reinforces the concentration pattern observed at the ministry level. The top ten organizations in the dataset account for a substantial share of total procurement value, while the long tail of remaining organizations contributes comparatively modest individual amounts.



This concentration has two practical implications for procurement governance. First, monitoring effort is most efficiently allocated toward the top-tier organizations, since they represent the greatest financial exposure. Second, any process improvement — such as faster payment cycles or stricter contract review — will have outsized impact if piloted with these organizations first.

11. Buyer Performance

Individual procurement officers (buyers) vary considerably in the volume and value of contracts they manage. A small number of buyers are responsible for handling a disproportionate share of total procurement value and contract count.



11.1 Procurement Concentration among Buyers

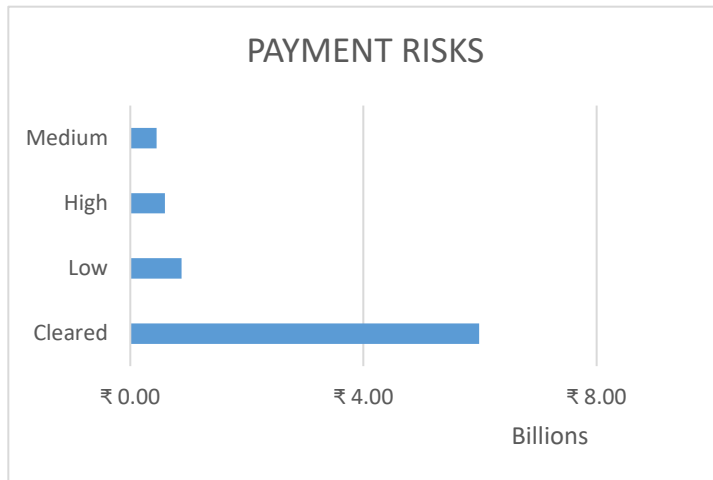
The top-ranked buyers by procurement value collectively manage a meaningful share of total government spend captured in the dataset. This concentration reflects the reality that procurement responsibility, like organizational spend, tends to cluster around a limited set of experienced officers rather than being distributed evenly.

11.2 Workload and Payment Responsibility

Buyers with the highest contract counts are not always the same individuals with the highest outstanding payment values, which suggests that payment delays are more closely tied to organizational process than to any single buyer's workload. This distinction matters for any future root-cause analysis of payment bottlenecks.

12. Risk Assessment

The dashboard includes a rule-based Payment Risk classification designed to flag contracts and organizations that warrant closer review, based primarily on the size of outstanding payment relative to contract value and billing stage.



12.1 High-Risk Contracts

Contracts flagged as high risk are concentrated among a relatively small number of organizations, reinforcing the broader concentration pattern seen throughout this report. These contracts represent the highest-priority targets for procurement and finance teams seeking to reduce outstanding liability.

12.2 Mega-Order Exposure

Mega-category contracts, while few in number, carry outsized financial exposure. A delay or dispute on a single mega contract can materially move the dashboard's aggregate pending-payment figure, which is why these contracts merit individual tracking rather than being managed only in aggregate.

12.3 Current Limitations of the Risk Model

The current risk classification is rule-based rather than predictive — it flags risk based on observed thresholds rather than forecasting which contracts are likely to become high risk. Section 16 outlines how this could evolve into a predictive model.

13. Executive Insights

The following insights synthesize the analysis above into concise, management-facing statements — the standard by which analyst work is typically judged in consulting and corporate analytics functions.

Insight 1 — Concentration of Spend. A small number of ministries and organizations account for a disproportionate share of total procurement value, indicating that government spending is structurally concentrated rather than evenly distributed.

Insight 2 — Clustered Outstanding Liability. Pending payments are concentrated within a limited set of organizations, meaning targeted intervention with this subset would resolve the majority of the government's outstanding payment exposure.

Insight 3 — Contract Size Imbalance. Small and Medium contracts dominate procurement activity by count, while Mega contracts contribute a disproportionate share of total procurement value — a classic Pareto pattern with direct implications for audit and review prioritization.

Insight 4 — Year-over-Year Variability. Procurement activity shows noticeable variation across years, reflecting shifting departmental priorities, budget cycles, and possibly external factors such as policy or fiscal-year changes.

Insight 5 — Buyer Concentration. A limited number of procurement officers manage a substantial share of total procurement value, indicating operational reliance on a small group of experienced buyers.

Insight 6 — Healthy but Uneven Payment Performance. An aggregate payment completion rate of 84.2% reflects reasonably strong overall performance, but this average conceals meaningfully lower completion rates within specific organizations.

Insight 7 — Early-Warning Value of Bill Status. The distribution of contracts across billing stages functions as a leading indicator — a growing share of contracts stalled in early stages would signal an emerging payment bottleneck before it appears in the pending-payment total.

14. Strategic Recommendations

The following recommendations translate the insights above into specific, actionable steps a procurement or finance function could take.

- Prioritize review of the organizations identified in Section 9.3 as carrying the largest share of outstanding payments, rather than applying uniform review across all organizations.
- Introduce automated payment reminders and escalation triggers for contracts that remain in early billing stages beyond a defined threshold.
- Establish dedicated governance and approval workflows for Mega-category contracts, given their disproportionate impact on total financial exposure.
- Develop standardized procurement performance benchmarks by organization, enabling like-for-like comparison rather than relying solely on aggregate totals.
- Extend the current rule-based risk model toward a predictive approach, using historical payment-delay patterns to flag at-risk contracts earlier in their lifecycle.
- Integrate the dashboard with a live or near-real-time GeM data feed to move from periodic snapshot reporting toward continuous procurement monitoring.

15. Limitations

This report and the underlying dashboard should be read with the following limitations in mind.

- The dataset is a static snapshot (06 July 2026) rather than a live transactional feed, so figures reflect a single point in time rather than continuous monitoring.
- Payment risk is currently determined through fixed, rule-based thresholds rather than a statistically validated predictive model.
- The dashboard is implemented in Excel, which supports strong interactivity for a single user but has limitations around multi-user collaboration and automated refresh compared to a dedicated BI platform.
- The dataset does not currently include vendor-side performance data (e.g., delivery timeliness, quality metrics), which limits analysis to the buyer / government side of each transaction.

16. Future Enhancements

This project is scoped as Version 1.0 of an evolving analytics product. The following roadmap outlines planned enhancements for subsequent versions.

Version	Enhancement
v1.1	SQL Server backend with a fully reproducible schema and query library covering executive KPIs, procurement, payment, buyer, and risk analysis
v2.0	Migration to Power BI with DAX measures, enabling richer interactivity and native cloud sharing
v3.0	Python-based ETL automation for ingesting new GeM exports without manual rework
v3.1	Predictive payment-delay model built on historical billing-stage transition data
v3.2	Vendor / organization performance scoring incorporating delivery and payment-cycle metrics
v4.0	Real-time procurement monitoring integrated directly with the GeM platform

17. Technical Architecture

The project follows a layered analytics architecture, moving from raw source data to executive-ready insight.

GeM Procurement Data → Excel Data Cleaning → Feature Engineering → Pivot Tables → Interactive Dashboard → SQL Server Analysis → Executive Insights

17.1 Excel Layer

The foundation of the project is a structured Excel Table containing the cleaned, feature-engineered procurement dataset. Pivot Tables and Pivot Charts are built directly on this table, with slicers providing interactive filtering by Ministry, Department, Order Year, Bill Status, and Order Size.

17.2 SQL Server Layer

The same structured dataset is designed to be loaded into a SQL Server database named GovernmentProcurementAnalytics, using a reproducible, script-driven setup rather than a manual import wizard. The SQL layer is organized into seven modules, each addressing a specific business domain:

Script	Business Domain
01_database_setup.sql	Database and table creation, bulk data load, and data-quality validation
02_executive_kpis.sql	Total procurement, amount paid, pending payment, completion rate, contract count
03_procurement_analysis.sql	Ministry, department, and organization procurement rankings; order-size segmentation; yearly trend
04_payment_analysis.sql	Pending payment by ministry, bill status distribution, top outstanding organizations, completion by year
05_buyer_analysis.sql	Top buyers by procurement value, pending payments, and contract volume
06_risk_analysis.sql	Payment risk distribution, high-risk contract listing, mega-order exposure
07_business_insights.sql	Top ministry, top outstanding organization, top buyer, and peak procurement year

A planned eighth script, 08_advanced_analysis.sql, extends the query library with Common Table Expressions, window functions (RANK, DENSE_RANK, ROW_NUMBER), running totals, year-over-year growth calculations, and Pareto-style concentration analysis — the class of query most commonly used to assess analytical depth in interviews for analyst roles.

18. Conclusion

The Government Procurement Intelligence Dashboard demonstrates a complete analytics workflow: transforming a raw, 46,738-record procurement export into a governed dataset, engineering the features required for meaningful analysis, and presenting the results through an executive dashboard designed to be understood within fifteen seconds.

The analysis surfaces a consistent theme across every dimension examined — ministries, organizations, buyers, and contract sizes — government procurement expenditure and payment activity are concentrated rather than evenly distributed. This concentration is not a flaw in the data; it is itself an actionable finding, since it means that a relatively small set of targeted interventions could materially improve payment performance and reduce outstanding liability across the portfolio.

Version 1.0 of this project establishes the Excel-based analytics layer described in this report. The roadmap in Section 16 outlines a clear path toward SQL Server integration, Power BI migration, and eventually predictive, real-time procurement monitoring — positioning this as an evolving analytics product rather than a one-time deliverable.